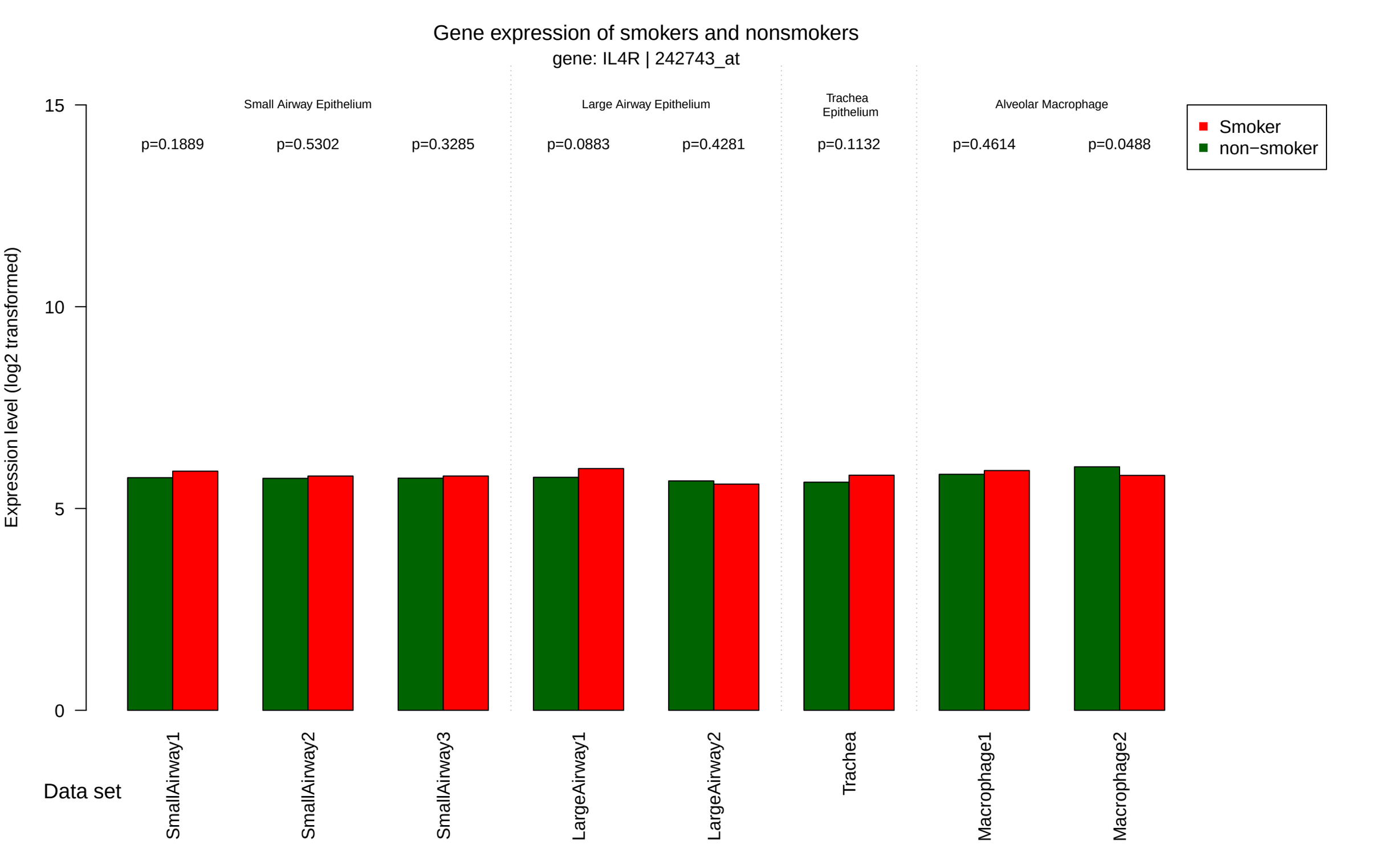
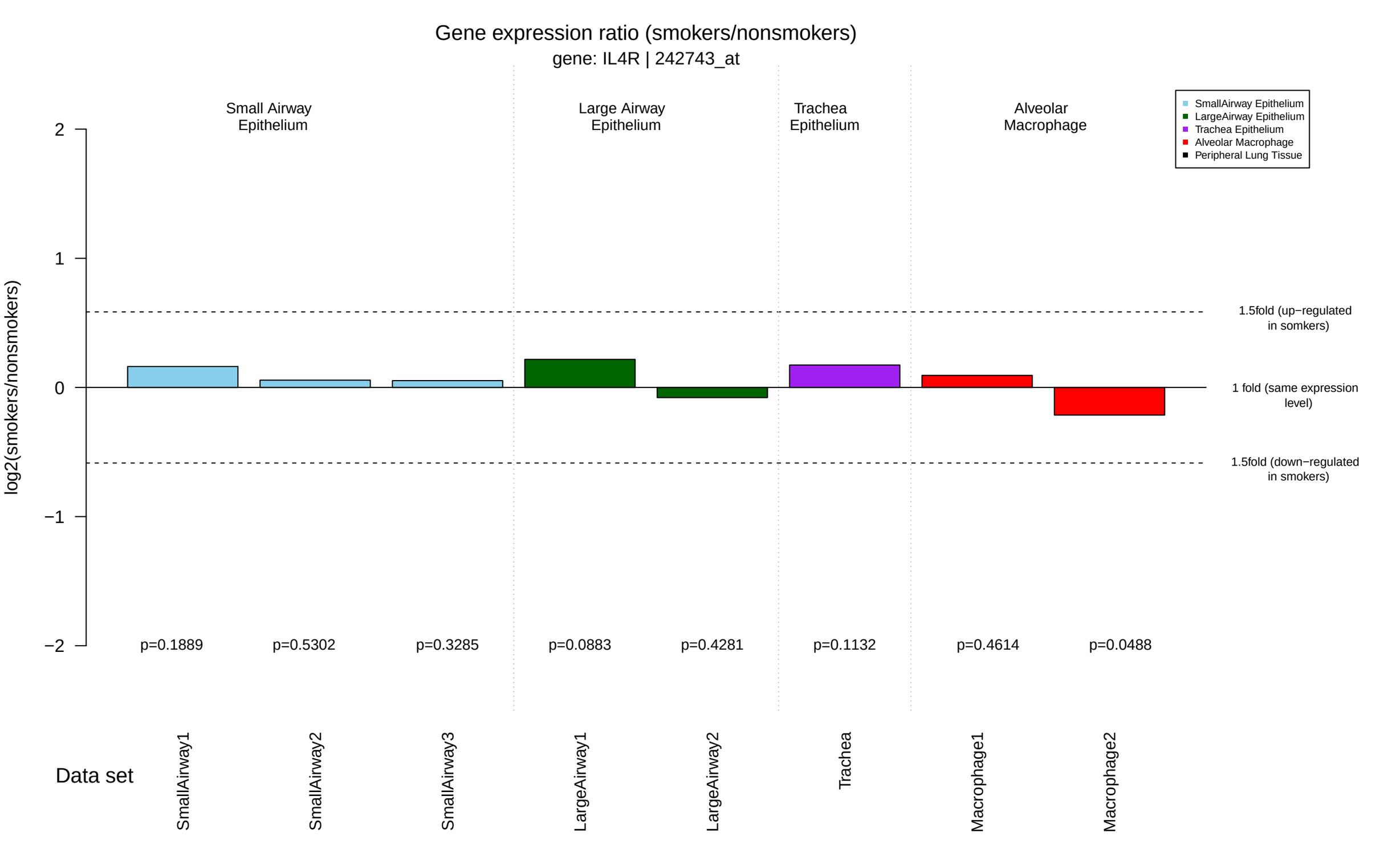
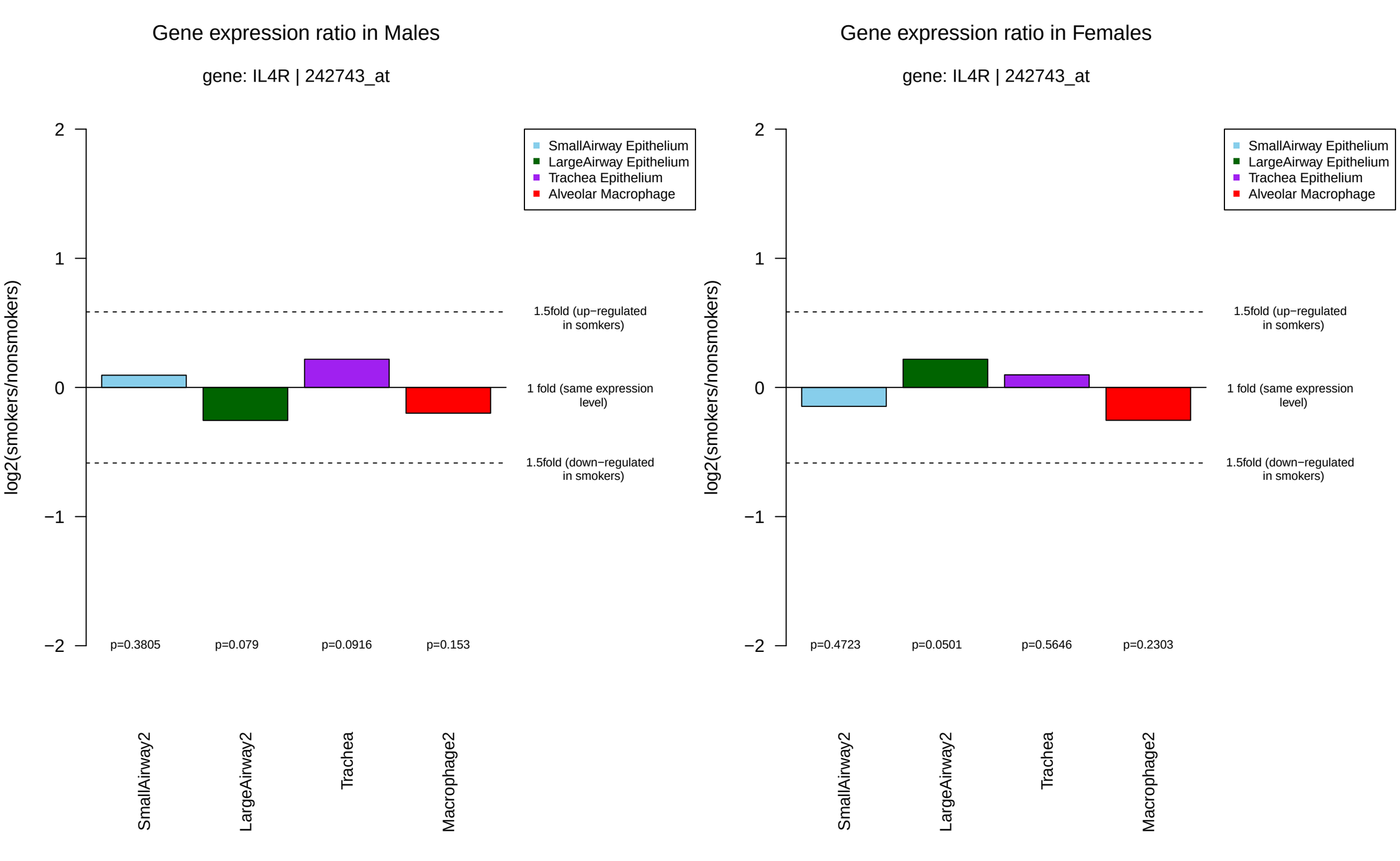
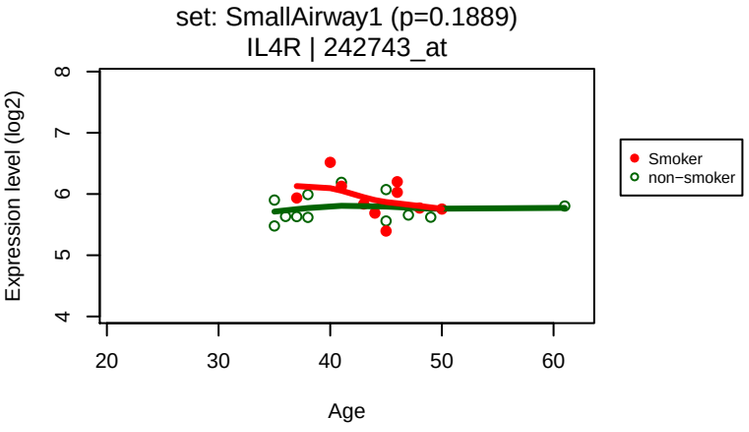
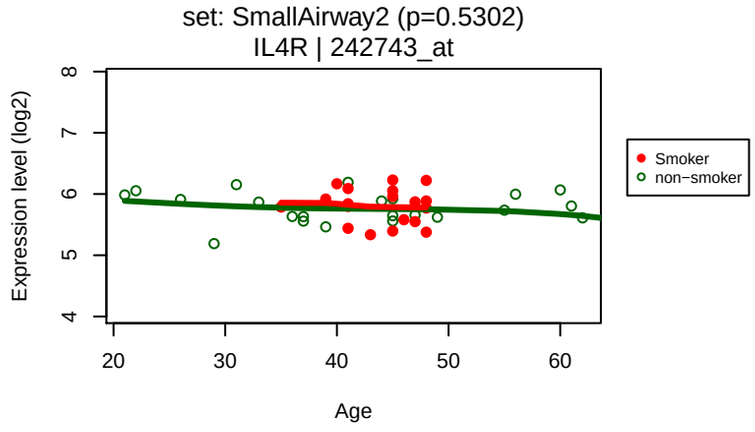
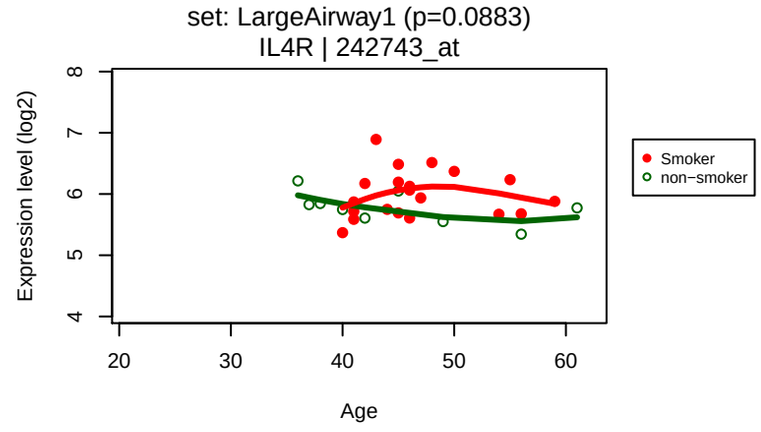

Gene expression of smokers and nonsmokers
gene: IL4R | 242743_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1889
p=0.5302
p=0.3285
p=0.0883
p=0.4281
p=0.1132
p=0.4614
p=0.0488
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

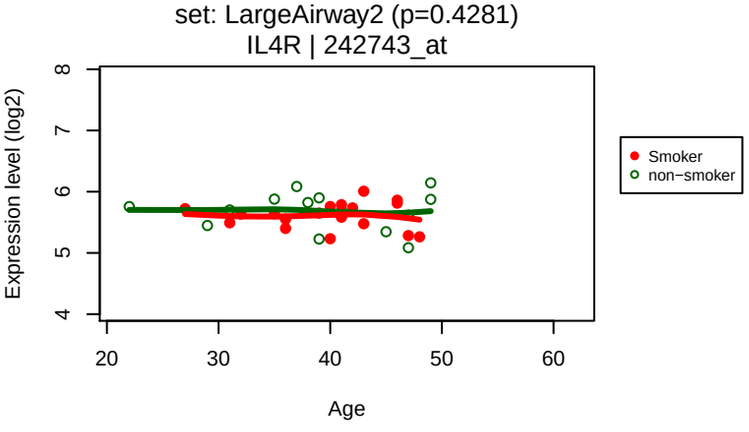

Gene expression ratio (smokers/nonsmokers)
gene: IL4R | 242743_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1889
p=0.5302
p=0.3285
p=0.0883
p=0.4281
p=0.1132
p=0.4614
p=0.0488
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

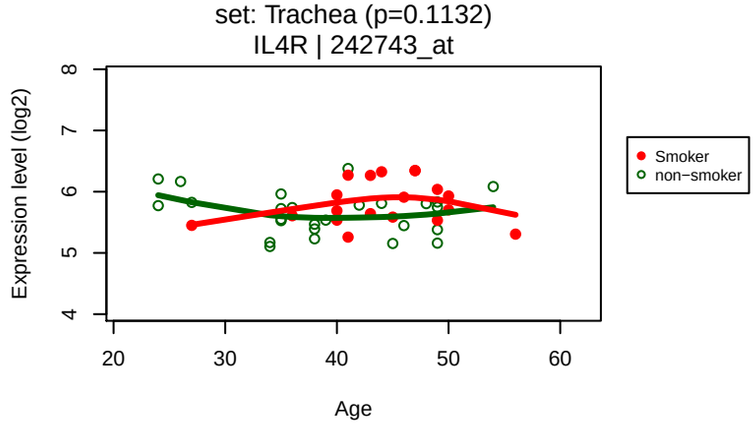

Gene expression ratio in Males
gene: IL4R | 242743_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3805
p=0.079
p=0.0916
p=0.153
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL4R | 242743_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4723
p=0.0501
p=0.5646
p=0.2303
SmallAirway2
LargeAirway2
Trachea
Macrophage2

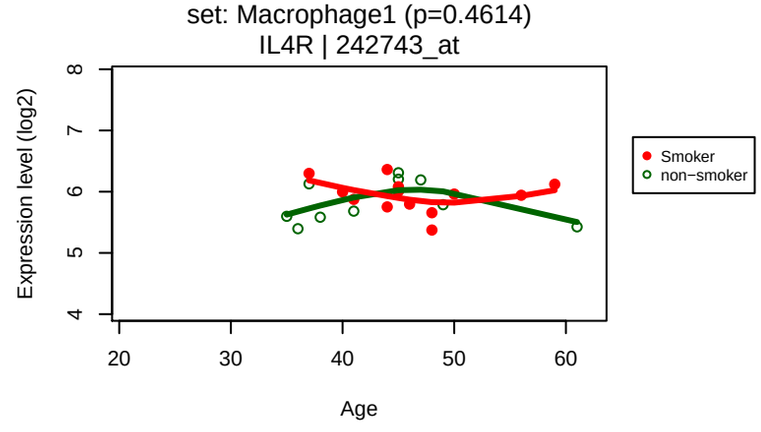

set: SmallAirway1 (p=0.1889)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker

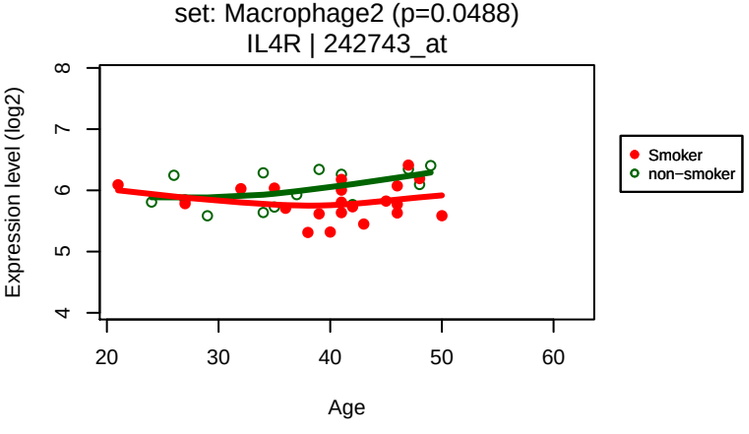

set: SmallAirway2 (p=0.5302)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker

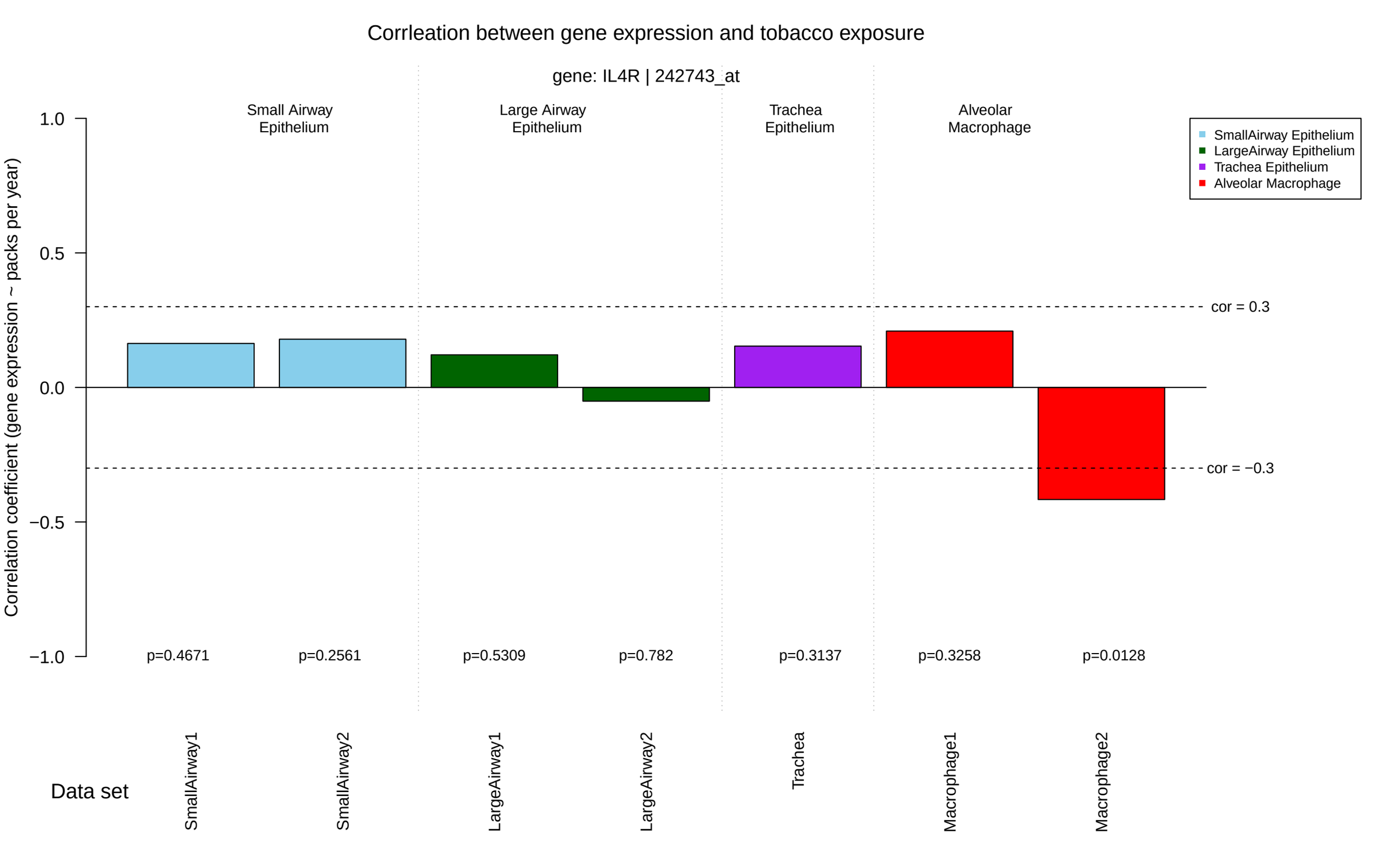

set: LargeAirway1 (p=0.0883)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4281)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1132)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4614)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0488)
IL4R | 242743_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL4R | 242743_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4671
p=0.2561
p=0.5309
p=0.782
p=0.3137
p=0.3258
p=0.0128
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2